

Extensive Exploratory Data Analysis

Project Mid Deliverable submission is due on 5th May, 2026. All students are required to make the submission.

Workflow

1. Choose a theme of interest and search relevant dataset(s).
2. Prior to beginning any visual analysis, write down at least three overall questions you would like to investigate for your chosen theme. For each question, briefly jot down your motivation for posing the question. For instance, was there something from the background knowledge that prompted you to ask this question? Or did you think of this question based on your experiences or stories you've heard from your friends and family? Etc. Cite these sources of evidence for your motivation.
3. Next, perform an exploratory analysis of the dataset through python notebook (any tool is acceptable). You should work through two different phases of exploration:
 1. In the first phase, you should seek to gain an overview of the shape & structure of the dataset. What variables does the dataset contain? How are they distributed? Are there any notable data quality issues? Are there any surprising relationships among the variables? Make sure to also perform "gut checks" for patterns you expect to see! Use map visualizations for geographical data.
 2. In the second phase, you should investigate each of your overall questions. For each question, start by creating a visualization that might provide a useful answer. Subsequently, refine the visualization (e.g., by adding additional variables, changing sorting or axis scales, transforming your data by filtering or subsetting it, etc.) to develop better perspectives, explore unexpected observations, or sanity check your assumptions.

Deliverable

Following deliverables should be submitted as part of Project Mid phase.

1. Properly formatted notebook with the entire data analysis, working and results. The file should be named as "A2_2023xxx_2023xxx.ipynb". This should be submitted as Assignment 2. (This is also evaluated as part of project.)

2. Presentation slides covering the entire data analysis, working and results. The file should be named as "PresentationProjectMid_2023xxx_2023xxx.ppt". This should be submitted as Project Mid.

Each member should present.

Rubric:

Component	Excellent	Satisfactory	Needs Improvement
Data Overview & Quality	A thorough overview of the data was achieved, with extensive profiling of fields and records to assess data quality.	Simple checks were conducted on only a handful of fields or records.	Little or no evidence that data quality was assessed.
Breadth of Exploration & Cited Motivation	Interesting overall questions target substantially different portions/aspects of the data. Rich motivation is well-grounded in multiple pieces of evidence that cite background readings, lived experiences, etc.	A nice set of overall questions are posed, but there is some overlap. Most questions are well-motivated, but there is an occasional gap in cited evidence.	Fewer than 3 overall questions were posed of the data, or there is significant conceptual overlap between them. Only shallow motivation is described, with little evidence cited.
Depth of Exploration	A sufficient number of follow-up questions were asked to yield rich insights that provide a robust understanding of the overall questions.	Some follow-up questions were asked and the seeds of insightful analysis planted, but further work could have pushed the analysis more deeply.	Few, if any, follow-up questions were asked after answering the overall questions. Only shallow analysis was conducted with any follow-up questions.

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Data Transformation	More advanced transformation were used to extend the dataset in interesting or useful ways.	Simple transforms (e.g., sorting, filtering) were primarily used.	The raw dataset was used directly, with little to no additional transformation.
Visual Expressiveness & Variation	No visualizations violate the expressiveness criteria, and a rich diversity of marks and visual encodings have been explored.	No visualizations violate the expressiveness criteria, but a greater variety of design choices (especially marks and encoding channels) might have helped spur further analysis.	Several visualizations violate the expressiveness criteria
Captions	Captions richly describe the visualizations and contextualize the insight within the overall analysis.	Captions do a good job describing the visualizations, but could better connect prior or subsequent steps of the analysis.	Captions are missing, overly brief, or shallow in their analysis of visualizations.
Creativity & Originality	You exceeded the parameters of the assignment, with original insights or particularly engaging visualizations.	You met all the parameters of the assignment.	-
In class Presentation	Your presentation is crisp, perhaps even exciting! Narration is clear, flows smoothly, and is easy for the audience to follow. Slides are well-constructed with a	The presentation is clear, and we understand the goals of your project. However, some presentation issues around the format, narration, or slide design hinder easy	Your presentation is difficult to follow and/or doesn't hold together. You may have gone over the time limit. Narration rambles and/or is difficult to follow. Slides are text-heavy, making poor or

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	balance of visual and textual information. Your presentation fits within the allotted 5 minutes.	comprehension. Your presentation fits within the allotted 5 minutes.	inadequate use of images or visual media. Or, conversely, you have visual-only slides without any text to help explain what the audience is seeing. Few to no members of the team showed up for the presentation.

Evaluation is not limited to above aspects. Projects may also be evaluated on overall creativity, visualizations, quality and qualitative aspects. All operational guidelines such as project and group submission in time, naming file properly, and others should be followed. Marks may be deducted if these guidelines are not followed.

Reference

<https://vis-society.github.io/>